

DAPRIL

(Lisinopril)

COMPOSITION

Each tablet contains either 5mg, 10mg or 20mg lisinopril as lisinopril dihydrate.

PHARMACEUTICAL FORM

DAPRIL 5mg tablets are peach, round, convex tablets, 6mm in diameter, for oral administration.

DAPRIL 10mg tablets are peach, round, convex tablets, 7mm in diameter, for oral

administration. DAPRIL 20mg tablets are peach, round, convex tablets, 8mm in diameter, for oral administration.

INDICATIONS

Hypertension: DAPRIL is indicated in the treatment of essential hypertension and in renovascular hypertension. It may be used alone or concomitantly with other classes of antihypertensive agents.

Congestive Heart Failure: DAPRIL is indicated in the management of congestive heart failure as an adjunctive treatment with diuretics and where appropriate, digitalis.

Acute Myocardial Infarction: DAPRIL is indicated for the treatment of haemodynamically stable patients within 24 hours of an acute myocardial infarction, to prevent the subsequent development of left ventricular dysfunction or heart failure and to improve survival. Patients should receive, as appropriate, the standard recommended treatment such as thrombolytics, aspirin and beta-blocker.

DOSAGE AND METHOD OF ADMINISTRATION

Tablets for oral administration taken once daily at approximately the same time every day. Absorption of Lisinopril is not affected by food.

The dose is individual for each patient according to his/her profile and blood pressure response.

Adults:

Hypertension: The usual recommended starting dose is 10mg once a day. Patients with renovascular hypertension, volume depletion, heart failure or severe hypertension, may experience an excessive fall of blood pressure after taking the initial dose and therefore in such patients an initial dose of 2.5-5mg is recommended. Initiation of treatment should take place under medical supervision. Maintenance dose, following careful dose titration upwards, if required, is usually 20mg once a day. Generally if the desired therapeutic effect is not achieved within 2 to 4 weeks, the dose can be further increased. The maximum daily dose is 80mg.

Patients treated with diuretics: Symptomatic hypotension may occur after initial therapy with DAPRIL in currently diuretic-treated patients. Caution is recommended therefore, since these patients may be volume and/or salt depleted. If possible, the patient should stop taking the diuretic 2-3 days before initiation of therapy with DAPRIL. In case that this is not feasible, a starting dose of 5mg is recommended. Renal function and serum potassium should be monitored. The subsequent dosage of DAPRIL should be adjusted according to blood pressure response. If necessary, diuretic therapy may be resumed.

Dosage adjustment in impaired renal function: Use with caution in patients with impaired renal function, dose reduction may be required. If creatinine clearance is less than 10ml/minute (including dialysis patients), initiate dose with 2.5mg. If creatinine clearance is 10-30ml/minute starting dose is 2.5-5mg. If creatinine clearance is 31-80ml/minute starting dose is 5-10mg. The maximum daily dose should not be more than 40mg. Dosage/frequency of administration should be adjusted to the blood pressure response.

Heart failure: Starting dose is 2.5mg with subsequent dose titration (i.e. by increments of no more than 10mg and at intervals of no less than 2 weeks) based on the clinical response of each individual. It should be initiated under close medical supervision. In the case of severe or unstable congestive heart failure, or in high risk patients such as those over 70, on high dose loop diuretics, multiple diuretic therapies, therapy should be initiated in hospital. If possible, diuretic dose should be reduced before beginning treatment. Blood pressure, serum potassium and renal function should be monitored before and during treatment. The usual maintenance dose, dependent upon patient response, is between 5mg and 35mg.

Acute myocardial infarction: Treatment with DAPRIL may be started within 24 hours of the onset of symptoms. Treatment should not be started if systolic blood pressure is lower than 100 mmHg. The first dose of DAPRIL is

5mg given orally, followed by 5mg after 24 hours, 10mg after 48 hours and then 10mg once daily thereafter. Patients with a low systolic blood pressure (120 mmHg or less) should be given a lower dose – 2.5mg orally- as initial treatment or during the first 3 days after the infarction.

In cases of renal impairment (i.e. creatinine clearance < 80ml/min), the starting dose of DAPRIL should be adjusted according to the patient's creatinine clearance. Maintenance dose is 10mg once a day. If hypotension occurs (systolic blood pressure less than or equal to 100 mmHg), a daily maintenance dose of 5mg may be given with temporary reductions to 2.5mg if needed. If prolonged hypotension occurs (systolic blood pressure less than 90 mmHg for more than 1 hour), DAPRIL should be withdrawn.

Treatment should continue for six weeks and then the patient should be re-evaluated. Patients who develop symptoms of heart failure should continue with DAPRIL (see *Dosage and administration* for heart failure). DAPRIL is compatible with intravenous or transdermal glyceryl trinitrate.

In the case of renal impairment, the starting dose of DAPRIL should be adjusted according to the patient's creatinine clearance, as described in detail, in paragraph *Dosage adjustment in impaired renal function*.

Use in kidney- transplanted patients: Not recommended.

Elderly: No age-related changes in the efficacy or safety profile of lisinopril have been reported and therefore no dosage reduction is usually necessary, unless renal function is impaired. In this case, the dosage described in paragraph *Dosage adjustment in impaired renal function*, should be followed and thereafter adjusted according to the blood pressure response.

Children: Not recommended

CONTRA-INDICATIONS

Pregnancy. Treatment should be stopped if pregnancy is suspected (see Pregnancy and lactation).

Lactation period.

Hypersensitivity to lisinopril.

Patients with a history of angioneurotic oedema in relationship to previous ACE inhibitor therapy.

Hereditary/idiopathic angioneurotic oedema.

WARNINGS AND PRECAUTIONS

Hypotension: It is more likely to occur in patients who have been volume depleted by diuretic therapy, dietary salt restriction, dialysis, diarrhoea or vomiting mainly in patients with severe heart failure. This is more likely in patients on high doses of loop diuretics, or those with hyponatraemia or functional renal impairment. In these patients treatment should be initiated under very close medical supervision. Diuretic treatment should be discontinued temporarily if possible. Such considerations also apply to patients with angina pectoris or cerebrovascular disease in whom excessive hypotension could result in myocardial infarction or cerebrovascular accident.

If hypotension develops, patients should be placed in the supine position. Volume repletion with intravenous normal saline may be required. The appearance of hypotension after the initial dose does not preclude subsequent careful dose titration with lisinopril after effective management.

Patients with renovascular hypertension: There is an increased risk of severe hypotension and renal insufficiency when patients with renovascular hypertension and pre-existing bilateral renal artery stenosis or stenosis of the artery to a solitary kidney are treated with lisinopril. Diuretic treatment may be a contributing factor. Loss of renal function may occur with only mild changes in serum creatinine even in patients with unilateral renal artery stenosis. In these patients treatment should be initiated under close medical supervision with low doses and careful dose titration. Diuretic therapy should be discontinued and renal function monitored during the first weeks of therapy.

Patients with renal insufficiency: Lisinopril should be used with caution in patients with renal insufficiency as they may require reduced or less frequent doses. Close monitoring of renal function during therapy should be performed as deemed appropriate in those with renal insufficiency. Renal failure has been reported in association with lisinopril therapy, mainly in patients with severe heart failure or underlying renal disease, including renal

artery stenosis. Some patients, with no apparent underlying disease have developed increases in blood urea and creatinine concentrations with concomitant diuretic administration. Dosage reduction of lisinopril and/or discontinuation of the diuretic may be required. It is recommended that renal function be monitored during the first weeks of therapy.

In acute myocardial infarction: Treatment with lisinopril should not be initiated in patients with evidence of renal dysfunction, defined as serum creatinine concentration exceeding 177 µmol/l and or proteinuria exceeding 500 mg/24 h. If renal dysfunction develops during treatment with DAPRIL (serum creatinine concentrations exceeding 265 µmol/l or a doubling from the pre-treatment value) then the physician should consider withdrawal of DAPRIL.

Haemodialysis patients: Patients who are dialysed using high flux polyacrylonitrile membranes and treated with lisinopril are likely to experience anaphylactoid reactions within a few minutes of starting haemodialysis. It is recommended to use an alternative membrane or an alternative category of antihypertensive drug.

Angioedema: may occur in patients treated with lisinopril or other ACE inhibitors, especially during the first weeks of treatment. However in rare cases severe angioedema may develop after long term treatment with an ACE inhibitor. Treatment should promptly be discontinued and replaced by an agent of another drug class.

Angioedema involving the tongue, glottis or larynx may be fatal. Emergency therapy should be given, including but not limited to, immediate subcutaneous adrenaline/epinephrine solution or slow intravenous adrenaline with control of ECG and blood pressure. Patients should be hospitalised and observed for at least 12 - 24 hours and should not be discharged until complete resolution of symptoms has occurred.

Cough: During treatment with ACE inhibitors, a dry and non-productive cough may occur which disappears after discontinuation.

Patients with liver insufficiency: Dosage adjustment is not necessary.

Elderly: Some elderly patients may be more responsive to lisinopril than younger patients. Administration of low initial doses and the evaluation of renal function at the beginning of treatment is recommended.

Children: There are no safety or efficacy studies of lisinopril in children. Use is not recommended.

Hyperkalaemia: Hyperkalaemia may occur during treatment with lisinopril, especially in the presence of renal insufficiency and/or heart failure. Potassium supplements or potassium sparing diuretics are generally not recommended. If concomitant use of the above mentioned agents is deemed appropriate, they should be used with frequent monitoring of serum potassium.

Surgery/anaesthesia: Lisinopril may cause hypotension or even hypotensive shock in patients undergoing major surgery or during anaesthesia due to the enhancement of other hypotensive agents. If it is not possible to withhold lisinopril, volume management should be handled with care.

Aortic stenosis/Hypertrophic cardiomyopathy: Use with caution in patients with an obstruction in the outflow tract of the left ventricle.

Neutropenia/Agranulocytosis: The risk of neutropenia appears to be dose and type related and is dependent upon the patient's clinical status. It is rarely seen in uncomplicated patients but may occur in patients with some degree of renal impairment especially when it is associated with collagen vascular diseases. It is reversible after discontinuation of lisinopril.

Proteinuria: It may occur particularly in patients with existing renal function impairment or those on relatively high doses of lisinopril.

Increased risk of birth defects, fetal and neonatal morbidity and death when used throughout pregnancy
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INTERACTIONS WITH OTHER MEDICAMENTS AND OTHER FORMS OF INTERACTION

Concomitant administration of potassium sparing diuretics or potassium supplements is not recommended, lisinopril attenuates diuretic induced potassium loss. Potassium sparing diuretics, e.g spironolactone, triamterene, or amiloride, potassium supplements or potassium containing salt substitutes may lead to significant increases in serum potassium. If concomitant use is indicated because of demonstrated hypokalaemia they should be used with caution and frequent monitoring of serum potassium levels.

The following precautions in use should be followed:

Patients on diuretics and especially those who are volume and/or salt depleted, may experience an excessive reduction of blood pressure after initiation of lisinopril therapy. The possibility of hypotensive effects can be reduced by discontinuation of the diuretic, increasing volume or salt intake prior to intake and by initiation of therapy with lower doses of lisinopril. Further increase in dosage should be with caution.

Concomitant administration of lithium with lisinopril may reduce the excretion of lithium. Serum lithium levels should be monitored frequently.

ACE inhibitors, including lisinopril, may enhance the hypotensive effects of certain anaesthetic drugs.

Postural hypotension may occur with concomitant administration of narcotics or anti-psychotics.

The hypotensive effect of lisinopril may be increased in concomitant administration with other antihypertensive agents.

Allopurinol, cytostatic or immunosuppressive agents, systemic corticosteroids or procainamide concomitantly administered with lisinopril may lead to an increased risk of leucopenia.

The following should be taken into account:

Administration of a non-steroidal anti-inflammatory agent may reduce the antihypertensive effect of lisinopril, and other ACE inhibitors. Furthermore, it has been described that NSAIDs and ACE inhibitors exert an additive effect on the increase in serum potassium, whereas renal function may decrease. These effects are in principle reversible, and occur especially in patients with compromised renal function.

Concomitant administration of antacids induces decreased bioavailability of ACE inhibitors.

Sympathomimetics may reduce the effect of ACE inhibitors; patients should be carefully monitored to confirm that the desired effect is being obtained.

Alcohol enhances the hypotensive effects.

The bioavailability of lisinopril is not affected by food.

PREGNANCY AND LACTATION

There are no appropriate and well controlled studies in humans. DAPRIL crosses the placenta and can cause foetal and neonatal morbidity and mortality when administered to pregnant women.

Foetal exposure to lisinopril during the second and third trimesters of pregnancy has been associated with neonatal hypotension, renal failure, face and skull deformities and/or death. Maternal oligohydramnios has also been reported, reflecting decreasing foetal renal function. Limb contractures, craniofacial deformities, hypoplastic lung development and intrauterine growth retardation have been reported in association with oligohydramnios. Infants exposed *in utero* to lisinopril should be closely observed for hypotension, oliguria and hyperkalaemia. Oliguria should be treated by support of blood pressure and renal perfusion.

Intrauterine growth retardation, prematurity, patent ductus arteriosus and foetal death have also been reported, but it is unclear if they were associated with lisinopril or the underlying maternal disease.

It is unknown if exposure limited to the first trimester can adversely affect foetal outcome. Women who become pregnant while receiving DAPRIL should be informed of the potential hazard to the foetus.

It is not known if DAPRIL is excreted in breast milk and the effect on the nursing infant is unknown. Breast-feeding while on DAPRIL treatment is not recommended.

Increased risk of birth defects, fetal and neonatal morbidity and death when used throughout pregnancy

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

There are no studies on the effect of DAPRIL on the ability to drive. When driving vehicles or operating machines it should be taken into account that occasional dizziness or weariness may occur.

UNDESIRABLE EFFECTS

The following side effects have been observed associated with ACE inhibitor therapy:

Cardiovascular system: Severe hypotension has occurred after initiation or increase of therapy. This occurs especially in certain risk groups (see Warnings and precautions for use). Symptoms like dizziness, feeling of weakness, impaired vision, rarely with disturbance of consciousness (syncope) can occur.

Individual cases of tachycardia, palpitations, arrhythmias, angina pectoris, myocardial infarction, transient ischaemic attacks and cerebral haemorrhage have been reported in association with hypotension.

Renal system: Renal insufficiency may occur or be intensified. Acute renal failure has been reported (see Warnings and precautions for use).

Respiratory system: ACE inhibitors have been documented to induce cough in a proportion of patients. Rarely dyspnoea, sinusitis, rhinitis, glossitis, bronchitis and bronchospasm have been reported. In individual cases angioneurotic oedema involving the upper airways has caused fatal airway obstruction.

Gastro-intestinal tract: Occasionally nausea, abdominal pain, indigestion, vomiting, diarrhoea, constipation and dry mouth can occur.

Individual cases of cholestatic icterus, hepatitis, pancreatitis and ileus have been described in relationship to therapy with ACE inhibitors.

Skin, vessels: Occasionally allergic and hypersensitivity reactions can occur like rash, pruritus, urticaria, erythema multiforme, Stevens-Johnson syndrome, toxic epidermic necrolysis, psoriasis and alopecia. This can be accompanied by fever, myalgia, arthralgia, eosinophilia and/or increased ANA titers. ACE inhibitors have been associated with the onset of angioneurotic oedema in a small subset of patients involving the face and oropharyngeal tissues.

Nervous system: Occasionally headaches, dizziness, weariness; rarely depressions, sleep disorders, paraesthesiae, impotence, disorders of balance, confusion, tinnitus, blurred vision and taste disorders.

Drug/Laboratory parameters: Increases in blood urea and plasma creatinine, reversible on discontinuation may occur, especially in the presence of renal insufficiency, severe heart failure and renovascular hypertension.

Decreases in haemoglobin, haematocrit, platelets and in white cell count, and in individual cases agranulocytosis or pancytopenia, as well as elevation of liver enzymes and serum bilirubin have been reported in a few patients. In patients with a congenital deficiency concerning G-6-PDH individual cases of haemolytic anaemia have been reported.

OVERDOSE

Symptoms of overdosage include severe hypotension, electrolyte disturbances and renal failure.

After ingestion of an overdose, the patient should be kept under close supervision, preferably in an intensive care unit. Serum electrolytes and creatinine should be monitored frequently. Therapeutic measures depend on the nature and severity of the symptoms. Measures to prevent absorption and hasten elimination should be applied such as gastric lavage, administration of adsorbents and sodium sulphate within 30 minutes after intake if ingestion is recent. If hypotension occurs, the patient should be placed in the shock position and salt and volume supplementation should be given rapidly. Treatment with angiotensin II should be considered. ACE inhibitors may be removed from circulation by haemodialysis. The use of high flux polyacrylonitrile membranes should be avoided.

PHARMACOLOGY

Pharmacodynamics: The beneficial effects of lisinopril in hypertension and heart failure appear to result primarily from the suppression of the plasma renin-angiotensin aldosterone system. Renin is an endogenous enzyme synthesised by the kidneys and released into the circulation where it converts angiotensinogen to angiotensin I a relatively inactive decapeptide. Angiotensin I is then converted by angiotensin converting enzyme, a peptidyl dipeptidase, to angiotensin II. Angiotensin II is a potent vasoconstrictor responsible for arterial vasoconstriction and increased blood pressure, as well as for stimulation of the adrenal gland to secrete aldosterone. Inhibition of ACE results in decreased plasma angiotensin II, which leads to decreased vasopressor

activity and to reduced aldosterone secretion. Although the latter decrease is small, small increases in serum potassium levels may occur, along with sodium and fluid loss. The cessation of the negative feedback of angiotensin II on the renin secretion results in an increase of the plasma renin activity.

Another function of the converting enzyme is to degrade the potent vasodepressive kinin peptide bradykinin to inactive metabolites. Therefore, inhibition of ACE results in an increased activity of circulating and local kallikrein-kinin-system which contributes to peripheral vasodilation by activating the prostaglandin system. It is possible that this mechanism is involved in the hypotensive effect of lisinopril and is responsible for certain side effects. In patients with hypertension, administration of lisinopril results in a reduction of supine and standing blood pressure to about the same extent with no compensatory increase of the heart rate. Peripheral arterial resistance is reduced with either no change or an increase in cardiac output.

There is an increase in renal blood flow and glomerular filtration rate is usually unchanged.

Achievement of optimal blood pressure reduction may require several weeks of therapy in some patients. The antihypertensive effects are maintained during long term therapy. Abrupt withdrawal of therapy has not been associated with a rapid increase in blood pressure.

ACE inhibitors are effective even in patients with low renin hypertension. Although antihypertensive effects have been found in the races studied, black hypertensive patients (usually a low renin hypertensive population) had a smaller average response to ACE inhibitor monotherapy than non-black patients. This difference disappears when a diuretic is added.

The haemodynamic effects of ACE therapy in patients with heart failure result from both arteriolar and venodilation. Systemic vascular resistance is decreased and venous capacity increased. Thus, pre and after load are reduced. Consequences are a decrease in left ventricular filling pressure/capillary wedge pressure and an increase in cardiac output; heart rate remains unchanged or may even decrease. Clinical signs and symptoms of the heart failure will improve and exercise capacity increase. These effects are maintained during long term treatment.

Pharmacokinetics: Peak serum concentrations appear within about 7 hours following oral administration of lisinopril. There is some evidence of a trend towards slight delay in reaching peak serum levels in patients with acute myocardial infarction. On multiple dosing, lisinopril has an effective half-life of accumulation of 12.6 hours. Declining serum concentrations show a prolonged terminal phase, although this does not contribute to lisinopril accumulation. The terminal phase probably represents saturable binding to ACE and is non proportional to dose. It would appear that lisinopril does not bind to other serum proteins.

Impaired renal function decreases the elimination of lisinopril, excreted by the kidneys, the decrease is only clinically important when the glomerular filtration rate is <30ml/min. Older patients have both higher plasma levels and higher area under the curve than younger patients. Lisinopril is removed by dialysis.

On data from urinary recovery, the mean extent of lisinopril absorption is about 25%, with a wide range of inter-patient variability, over dosages of 5mg to 60mg.

Lisinopril is not metabolised, and the absorbed drug is excreted unchanged in the urine.

Food does not affect the absorption of lisinopril. Lisinopril crosses the blood brain barrier poorly, as evidenced by studies in rats.

PRECLINICAL SAFETY DATA

No further relevant data.

INCOMPATIBILITIES

None known

STORAGE CONDITIONS

Store below 30°C in the original package

SHELF LIFE

Four (4) years

PHARMACEUTICAL PARTICULARS

List of Excipients: DAPRIL tablets also contain mannitol, dibasic calcium phosphate dihydrate, red iron oxide, maize starch, pregelatinised starch and magnesium stearate.

Nature and Contents of Container: Combination aluminium polyvinylchloride blisters of ten tablets. Packs of 30, 500 and 1000 tablets with a patient information leaflet are available.

Instruction for Use/Handling: Tablets should be swallowed whole, with a little water if required. They can be taken with or without food.

NAME AND ADDRESS OF MANUFACTURER

Dapril 5mg: Medochemie Ltd, Central Factory, 1 – 10, Constantinoupolis Street, Limassol 3011 Cyprus
Dapril 10mg, 20mg: Medochemie (Far East) Ltd, (Oral Facility), No. 40, VSIP II Street 6, Vietnam-Singapore
Industrial Park II, Binh Duong Industry-Service-Urban Complex, Hoa Phu ward, Thu Dau Mot city, Binh Duong
province, Vietnam

NAME AND ADDRESS PRODUCT REGISTRATION HOLDER

Komedic Sdn Bhd, 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor DE.

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